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Article in Journal of Travel Research · April 2024

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Abstract

Recent advancement in Generative Conversational Artificial Intelligence (AI) has opened the possibility of promoting chatbots for nudging. Tourists returning home from destinations that enforce sustainable transport regulations might be nudged to continue traveling sustainably. Through exploratory qualitative research, this study conceptualizes a chatbot to facilitate pro-environmental behavior spillover among domestic tourists in the Gili Islands, Indonesia. The Service Design approach and Grounded Theory were applied throughout the data collection process involving semi-structured in-depth interviews, focus group discussions and field observations. Insights obtained from stakeholders from multidisciplinary backgrounds were used to provide an overview of how the chatbot can be developed and the potential impact it could bring. A theoretical model predicting factors that play important roles in adopting the proposed chatbot and its respective nudges was developed. It informs policymakers and practitioners about the potential use of AI to help transform tourist experiences and shape lasting responsible behaviors.

Keywords

artificial intelligence, chatbot, nudging, pro-environmental behavior spillover, tourism

Introduction

Tourism is a carbon-intensive industry. It is estimated that the industry contributed to 11% of global greenhouse gas emissions in 2019 (Girma, 2022). Faced with challenges related to climate change, scholars have called for more research into making tourism more environmentally friendly (Scott & Gössling, 2022). Dolnicar (2020) proposed approaches designed to reduce the environmental harm caused by tourists, ranging from providing feedback on their resource use to reducing plate sizes to minimize food waste. While these approaches are useful on-site, more can be done to allow these pro-environmental behaviors (PEBs) to spill over from the tourism context into daily life. For instance, if tourists have performed PEBs during their visit to a particular tourist destination, they should be made aware of the positive impacts that have resulted from those behaviors (Van der Werff et al., 2014). Self-reflection on this information is expected to awaken their environmental consciousness and help them form lasting, responsible behaviors.

The above process can be explained by the concept of the transformative tourism experience, where tourists

often undergo a significant change after engaging in socially and environmentally responsible interactions with nature (Kirillova et al., 2017). Previous literature has provided evidence for the potential of transformative tourism to encourage short- and long-term behavior change, specifically related to PEB, and called for more research into this potential (e.g., Sheldon, 2020; Wolf et al., 2015, 2017). For a transformative tourism experience to occur, tourists must possess the ability to consciously derive meaning from their travel-nature encounters and engage in self-reflection regarding their past experiences (Kirillova et al., 2017). This is because tourists often take their experiences for granted unless a specific mechanism

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is introduced to facilitate the self-reflection process. Tourism providers, thus, should introduce service innovations to facilitate tourists' self-reflection and induce the transformation process (Fu et al., 2015).

The rapid advancement in AI innovations introduces the possibility that tourism providers will be able to reinvent how tourists enjoy their travel experiences. AI refers to the type of technology that can think or act in a humanlike and rational way (Tussyadiah, 2020). Chatbots, conversational agents that can interact with users through natural languages (Ling et al., 2021), are increasingly powered by AI. Past research has investigated the effective use of chatbots to promote positive changes in the behavior of users (Zhang et al., 2020). With the ever-increasing intelligence level of Generative Conversational AI, as shown in recent applications of large language models (LLMs), including ChatGPT (Dwivedi et al., 2023), it is increasingly relevant to discuss how chatbots can be used to facilitate self-reflection among tourists. This is aligned with recent conceptual studies on ChatGPT that advocate its use for personalizing customer experiences (e.g., Gursoy et al., 2023).

The sustainable transport regulations implemented on the Gili Islands in Lombok, Indonesia, condition tourists to travel sustainably throughout their trip to the islands (Olsson et al., 2022). No motorized vehicles, such as cars and motorcycles, are allowed on the islands (González-Rodríguez & Tussyadiah, 2021), giving tourists limited mobility options, including walking, cycling, and hiring traditional horse carriages called *cidomo*. These mobility options are part of sustainable travel behavior, a form of mobility that is energy-efficient and respectful to the environment (Olsson et al., 2022). Through a proper human-chatbot interaction, one can expect that chatbots may be able to assist tourists in their efforts to continue to behave pro-environmentally upon their return home by reducing their personal use of cars or motorcycles—demonstrating a PEB spillover effect. Therefore, further discussion of how such chatbots can be designed and developed is needed.

However, an idea about a new type of technological innovation should be accompanied by a detailed explanation of how the proposed solution will be effective. According to the Service Design principles, problems and opportunities should be framed in the right way to truly consider the needs of the user and the creator, instead of jumping straight to a “solution” (Stickdorn et al., 2018). For instance, existing research on chatbots designed to improve environmental sustainability is mostly focused on providing technical and practical solutions without strong theoretical contributions that can be used as guidance to replicate the proposed solutions to different sustainability problems in other contexts (Åberg, 2017; Gruber, 2022; Peniche-Avilés et al., 2016). This is contrary to the Service Design principles that emphasize the

importance of strong inquiry and exploration of the “how” and the “why” of the opportunity space in order to make true innovation possible, mostly using a range of qualitative research methods (Stickdorn et al., 2018).

This study formulates two research objectives to discuss the potential implications behind the proposed concept of a chatbot that facilitates PEB spillover among tourists. Firstly, it aims to conceptualize the chatbot technology and examine its potential implications. Secondly, the study seeks to identify enablers of and hindrances to PEB spillover effects, aiming to develop a theoretical model linking chatbot adoption and PEB spillover. To achieve these goals, this study employs an exploratory qualitative approach, including semi-structured in-depth interviews, focus groups and field observation, utilizing Service Design for data collection and Grounded Theory for analysis. The outcomes are expected to provide valuable insights for scholars and practitioners on chatbots' potential role in transformative tourism experiences.

Although focused on a specific context, the findings will be transferable to broader contexts due to their general theoretical insights. For example, post-visit action resources such as newsletters and electronic fact sheets that have been discussed in previous tourism literature as a means to facilitate PEB spillovers (e.g., Ballantyne et al., 2011; Hughes et al., 2011) can instead be delivered in the form of AI-powered chatbots. These resources contain additional learning sources developed based on research on environmental issues and behavior change (Bueddefeld & van Winkle, 2017). The findings of this study, which include the rich descriptions of the investigated phenomena and a theoretical model resulting from an analysis that follows a Grounded Theory approach (Wiesche et al., 2017), will assist the process of turning conventional post-visit action resources into conversational. As such, this study provides significant implications, particularly for policymakers, to leverage AI solutions for environmental challenges, using tourist destinations as a catalyst for extending PEBs.

Literature Review

Nudge Theory, Pro-Environmental Behavior and Spillover Effect

To enable PEB spillovers, interventions are needed so that people can change their behavior. Before proceeding with the development of interventions, one needs to justify the philosophy and theoretical underpinning that serve as the basis for designing the interventions. Previous literature has discussed a range of theoretical underpinnings to base the development of the interventions depending on the target behavior, such as boosting, shoving, and nudging (Oliver, 2015; Rouyard et al., 2022). Boosting focuses on giving people the skills or tools they need to change their

behavior, such as in the case of people following a regimen for a healthier life (Rouyard et al., 2022). Shoving represents an explicit regulation of individual behavior under the coercive paternalism philosophy, such as a ban on smoking (Oliver, 2015). While these two approaches can be effective in changing people's behavior, nudging may be considered the most appropriate theoretical basis for the context of people taking environmentally friendly transport since it preserves people's liberty while achieving the target behavior that can be considered less complex than a health regimen (Thaler & Sunstein, 2008).

Rooted in libertarian paternalism, nudge theory may inform the design of transformative tourism experiences aimed at influencing behavior. Nudging preserves choice freedom while encouraging better decisions through the given nudges (Thaler & Sunstein, 2008), which either stimulate intuitive thinking (e.g., evoking emotions, simplifying processes), foster reflexive thinking (e.g., reminders, reflective opportunities), or set default choices for automatic behavioral adjustments (Beshears & Gino, 2015). Nudges are particularly effective in promoting PEBs, which are actions consciously undertaken to minimize environmental impact (Kollmuss & Agyeman, 2002). The nudges can be designed to tap into and resonate with various value orientations, including biospheric, self-transcendent, social, egoistic, social-altruistic, biocentric, ecocentric and anthropocentric, all significant drivers of PEBs (Foroughi et al., 2022; Gatersleben, 2014; S. Schwartz, 2006; Stern et al., 1993; Thompson & Barton, 1994). Practical nudging examples include feedback on energy usage, default green electricity options or reduced plate sizes to curb food waste (Delmas et al., 2013; Kallbekken & Sælen, 2013; Pichert & Katsikopoulos, 2008).

In the tourism context, a recent systematic review on nudges for sustainability found that most current applications of nudges focus on the pre-trip and during-trip activities, such as towel reuse, energy conservation, carbon offsetting and conservation donation (Souza-Neto et al., 2022). The issue of the ways in which tourism can serve as a nudging mechanism that can impact the tourists' lives after the trip still receives limited scholarly attention. However, a notable work by Hehir et al. (2021) highlights tourism's potential to induce positive behavior changes among tourists that can shape responsible behavior. They traced the alumni of a youth polar expedition program and found that many continue to perform PEBs even years after their polar voyage, thanks to their perceived environmental self-identity.

The above phenomenon illustrates the concept of PEB spillovers. This refers to how performing certain PEBs can lead to the performance of the same or similar PEBs by individuals, thereby demonstrating a spillover effect (Van der Werff et al., 2014). Van der Werff et al. (2014) conducted an experiment that showed how strengthening a person's environmental self-identity by reminding them

of their past performance of PEBs can be the catalyst for the performance of the same or similar PEBs. Providing reminders is a form of nudging (Beshears & Gino, 2015), specifically nudging with information (Acquisti et al., 2017). In the tourism context, such a nudging mechanism can be implemented by reminding tourists of their past PEBs performed at certain tourist destinations and the positive impact that they have created from those PEBs. In the context of the present study, tourists who have traveled sustainably during their visit to the Gili Islands can be reminded of the environmental benefits of their past actions.

However, tourists traveling sustainably on the Gili Islands do not decide to take environmentally friendly transport because of their commitment to any of the values identified as the drivers of PEBs. Instead, they do so because the PEB is the mainstream behavior (Font & McCabe, 2017). Without any mechanism introduced to improve the tourists' environmental awareness (i.e., knowing the impact of their behavior on the environment), maintaining the performance of similar behavior outside the tourist destinations will be difficult (Wu et al., 2021), as no improvement is made to either one's biospheric or self-transcendent values (Kollmuss & Agyeman, 2002). As such, the introduction of certain nudging mechanisms to facilitate the PEB spillover is an important research topic that will shed light on how to best ensure that the positive experience gained from the tourism activity in Gili is internalized and spills over into the daily lives of the tourists.

Tourism providers can leverage conversational AI applications such as chatbots to nudge tourists with information to improve their environmental awareness (Majid et al., 2023). Chatbots offer advantages such as personalized cognitive engagement and 24/7 fast response time (Ling et al., 2021). In addition, much research has been conducted into the use of chatbots for lifestyle modification programs (Zhang et al., 2020). Such research can be categorized into pro-self-nudging, that is, when implementing the given nudges will benefit the individuals themselves (Hagman et al., 2015). Meanwhile, limited scholarly attention has been given to research into chatbots for pro-social nudging, where implementing the nudges may benefit others as well as the environment (e.g., Peniche-Avilés et al., 2016). In this study context, chatbots can be designed to facilitate PEB spillover among tourists who have visited nature-based destinations by providing them with sustainability-related nudges.

Technology Acceptance Theories and Frameworks for Chatbot-Based Nudging

Chatbots for pro-social nudging present an entirely different challenge than that presented by other chatbot

applications, such as those in the customer service context. The users of chatbots for pro-social nudging are required to follow the given nudges to meet the goal of changing their behavior so that it becomes more environmentally friendly. Thus, identifying factors that might hinder the user's adoption of the given pro-social nudges is crucial. In contrast, other chatbot applications and technologies traditionally center around how they can help solve the needs and wants of the users. Hence, such technologies can be analyzed sufficiently using existing technology acceptance theories such as the Technology Acceptance Model (TAM; Davis, 1989), the Unified Theory of Acceptance and Use of Technology (UTAUT; Venkatesh et al., 2003), and the Artificially Intelligent Device Use Acceptance model (AIDUA; Gursoy et al., 2019) and their derivatives.

For instance, in the context of chatbots for tourism, Pillai and Sivathanu (2020) extended TAM, while Hanji et al. (2023), Melián-González et al. (2019) and Zhang et al. (2022) extended UTAUT2 with a different set of additional factors to investigate the drivers for chatbot adoption. The theoretical frameworks offered by these studies are not applicable to the current study because they cannot fully explain the aforementioned particularities of chatbots designed to deliver pro-social nudges. Nevertheless, factors in these theoretical models, such as performance expectancy and effort expectancy, are predicted to still be relevant for the theoretical model in this study since using technology generally engenders user evaluation toward these two factors regardless of the context. Meanwhile, existing research on chatbots that deal with pro-social behavior such as donation mainly focuses on identifying the effects of individual factors like smile (Baek et al., 2022) and gratitude expression (Namkoong et al., 2023), or theories like moral judgment (Y. Zhou et al., 2022) and empathy (Park et al., 2023), rather than the holistic theoretical discussions of such chatbots as scientific phenomena. Furthermore, the framework proposed by Zhang et al. (2020) for the design of chatbots for changing human behavior is too practical to predict how people will respond to using the designed chatbot.

To conclude, the nudge theory serves as the theoretical grounding for exploring chatbot-based nudging in this study. Meanwhile, the PEB spillover theory is the foundation for formulating questions on how tourists can continue using environmentally friendly transport daily. Classic PEB literature suggests that "science and policy require a socioeconomic theory of behavior that incorporates both external conditions and internal processes" (Guagnano et al., 1995, p. 700). Therefore, a theoretical model that can map the factors that predict the adoption of the chatbot technology and its respective nudges is urgently needed to enable a greater understanding of the interaction between the above two theoretical concepts: chatbot-nudging and PEB spillover. Because AI-based

solutions to social and environmental problems are expected to flourish in the future, this study seeks to ensure that the creators of such innovations will be informed of any challenges embedded in such an endeavor.

Methods

Study Context and Proposed Technology

Investigating spillover effects that can emerge from tourism settings requires a concrete case study because we need to investigate how the exact same PEB that tourists perform in situ will be replicated in their daily lives (Wu et al., 2021). In a business-as-usual scenario, tourism does not engender PEBs. Instead, it facilitates holiday enjoyment that is fueled by hedonic motivations (Dolnicar, 2020). Therefore, finding a tourist destination where a real pro-environmental policy is being implemented to bring about PEBs among its tourists is required in order to base the proposed nudges. Without a concrete study context, proposing nudging mechanisms with a spillover orientation would be difficult. Lehner et al. (2016) provided several examples of nudging approaches without conducting a case study, which resulted in a less holistic overview of the studied phenomena. Bearing this in mind, the Gili Islands offer an appropriate context for this study as they implement a customary law oriented toward environmental sustainability. All tourists on the islands will have performed the same PEB before leaving, which allows them to receive nudges in the form of reminders of their past PEB.

The Gili Islands consist of three small islands (i.e., Gili Trawangan, Gili Meno, and Gili Air) located in the Northwest of Lombok, in the West Nusa Tenggara province of Indonesia. The total land area of these three islands is only 2.56 square miles, or around 6.65 km² (KKP, 2010). These islands were once remote destinations that offered peace to tourists worldwide who started coming in the early 1980s (Dickerson, 2008). However, intensive tourism development has occurred in the past decade, leading to a profusion of hotels and restaurants. Despite the rapid rate of development, the islands retain a customary law prohibiting the use of motorized vehicles such as cars and motorcycles. During high seasons, the Gili Islands can receive about 3,000 tourists each day (Halwi, 2023).

This study aims to conceptualize a chatbot technology that can facilitate PEB spillover among tourists on the Gili Islands. The initial technology concept was developed following iterative discussions amongst authors' professional networks for around 2 months to anchor the discussion with the research participants. The professional networks included international tourism academics, academics working in various social science fields (e.g.,

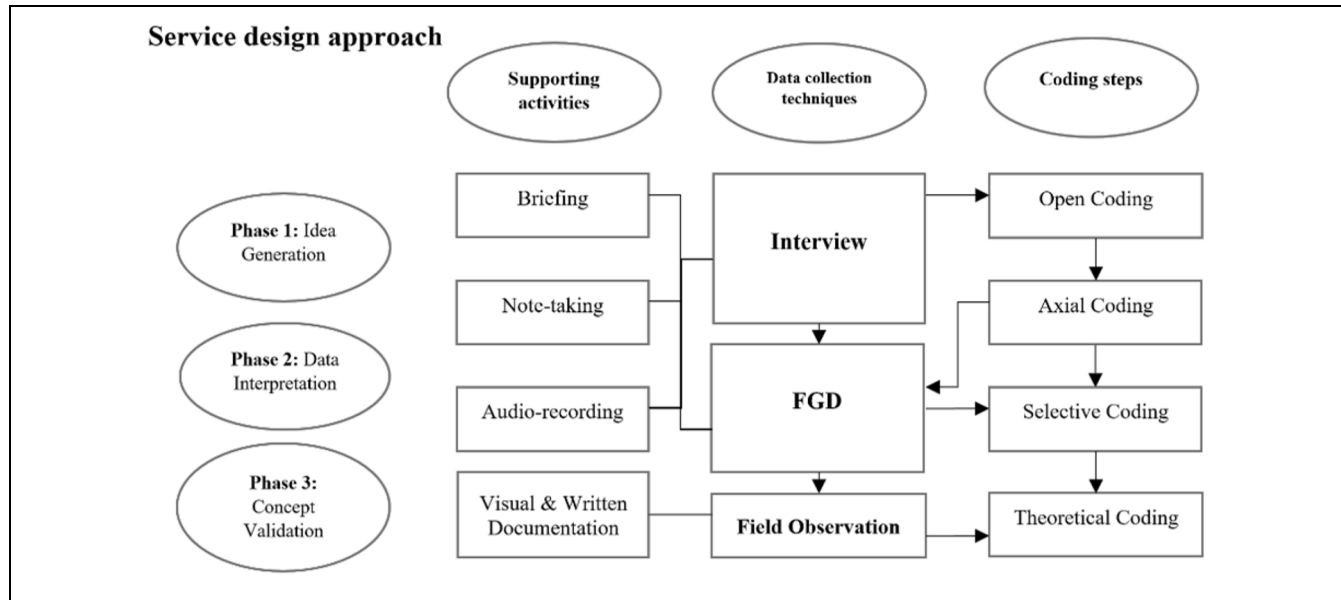


Figure 1. Data collection and analysis process.

psychology, transport, technology, linguistics) and individuals who had visited the Gili Islands as tourists—most of whom were Indonesian citizens. Once the concept was deemed workable, a prototype was developed with an explanation of how it would work, which is summarized in the following scenario:

To target pro-environmental behavior spillover, the chatbot will message tourists via WhatsApp once they have left the islands. The message will be about their experience on the islands, encouraging reflection on their sustainable travel behavior. It will be followed by educational information about the positive impact they have created by behaving pro-environmentally. The users can continue to chat with the chatbot to get more information.

Data Collection and Analysis

This study employed an exploratory qualitative approach (see Figure 1), combining three data collection techniques: semi-structured in-depth interviews, focus group discussions (FGDs), and participant observation (Creswell, 2007; Zhu et al., 2020). The Service Design approach guided the overall data collection process in order to conceptualize a new type of technology that is expected to solve a specific problem (Stickdorn et al., 2018). Following the Grounded Theory, the data collection and analysis were conducted iteratively (Creswell, 2007). Grounded Theory helps uncover and explain new phenomena through detailed descriptions of the new phenomena (i.e., narratives of empirical observations) and models (i.e., the definition of abstract variables and their relationships) (Wiesche et al., 2017).

Interviews were conducted to facilitate in-depth discussions with each participant so they could build on their responses (Saunders et al., 2019). All the participants were selected using theoretical sampling. According to Corbin and Strauss (1990), this sampling method requires the researchers to identify groups of individuals that are considered relevant to the studied phenomena and adjust the search for relevant participants based on the representativeness of the concepts found during the data collection process. This recommendation aligns with the “co-creative” principles from the Service Design approach, suggesting that potential key actors should be included in the service design processes, including the research and prototyping stages (Stickdorn et al., 2018).

To do this, the authors first analyzed information about any stakeholders who were considered relevant to the studied phenomena to map the general profile and expertise of the prospective participants. Thereafter, experts in chatbot technology and sustainable tourism were consulted on the proposed list. Following expert feedback, new stakeholders, including tourists who had visited the Gili Islands, were added. Once confirmed, the recruitment information was shared through the authors’ professional network. Participants were asked to sign a consent form before the interview. The interviews were concluded once theoretical saturation had been reached (Creswell, 2007). In total, 20 participants from different backgrounds were interviewed (see Table 1). In addition to three participants who had been invited to represent the perspectives of “tourists,” 10 more participants stated that they had visited the Gili Islands as tourists. Those who had been to Gili often used their past tourism experiences as a point of reference for their opinions. They

Table 1. List of Interview and FGD Participants.

ID	Gender	Age (years)	Education level	Monthly income	Years of experience	Role	Expertise	Domicile	Visited Gili?	FGD
Informants to inform technology and process development										
T1	Male	26–30	Bachelor	above \$1,345	4–8	Manager	Artificial intelligence, Data analyst	Jakarta ^a	V	V
T2	Male	26–30	Bachelor	\$800–1,345	2–4	Manager	Chatbot, Business Development, Marketing	Sleman ^b		V
T3	Male	31–35	Bachelor	above \$1,345	4–8	Founder & CEO	Chatbot, Computer Scientist	South Tangerang ^b		
T4	Male	31–35	Master	above \$1,345	8–15	Founder & CEO	Chatbot, Computer Scientist	Jakarta ^a	V	
T5	Female	36–40	Bachelor	above \$1,345	8–15	Professional	Natural Language Processing (NLP), Linguist	Majalengka ^c		
Informants to inform social change and sustainability preparedness										
S1	Male	36–40	Master	\$800–1,345	8–15	NGO Director	Government Contractor, Lecturer, Business Development	Tasikmalaya ^c	V	V
S2	Female	41–45	Doctor	\$800–1,345	15 +	Academic	Sustainability in Indonesia	Bandung ^a		V
S3	Male	26–30	Master	above \$1,345	0–2	Academic	Transportation in Indonesia	Sleman ^b	V	V
S4	Female	36–40	Master	\$200–335	8–15	Academic	Psychology	Yogyakarta ^a		V
S5	Female	36–40	Doctor	above \$1,345	8–15	Academic	Tourism, Destination Image	Bandung ^a		
S6	Female	31–35	Bachelor	\$800–1,345	4–8	NGO Manager	Sustainability activist in Gili from the UK	North Lombok ^c	V	
S7	Female	26–30	Master	above \$1,345	4–8	Consultant	Sustainable transport design and planning	Jakarta ^a		
Informants to inform destination management and experience										
D1	Male	31–35	Master	\$540–800	N/A	Tourist	N/A	Cikarang ^b	V	V
D2	Male	26–30	Senior high school	\$335–540	N/A	Tourist	N/A	Malang ^b	V	V
D3	Male	31–35	Senior high school	\$ 0–100	N/A	Tourist	N/A	Central Lombok ^c	V	V
D4	Female	41–45	Bachelor	\$335–540	2–4	Government official	Destination management of Gili Islands	North Lombok ^c	V	V
D5	Male	41–45	Master	\$540–800	8–15	Government official	Marketing of Gili Islands	North Lombok ^c	V	V
D6	Male	41–45	Senior high school	\$100–200	4–8	Manager	Tour organizer for Gili Islands experience	West Lombok ^c	V	V
D7	Male	41–45	Senior high school	\$200–335	2–4	NGO Manager	Environmental activist in Gili from local	North Lombok ^c	V	V
D8	Female	31–35	Bachelor	\$335–540	2–4	Government official	Ministry of Tourism	Jakarta ^a	V	

Note. The bigger the city is, the better the infrastructure quality and access to alternative transports are.

^aBig cities.

^bMedium-sized cities.

^cSmall cities/regencies.

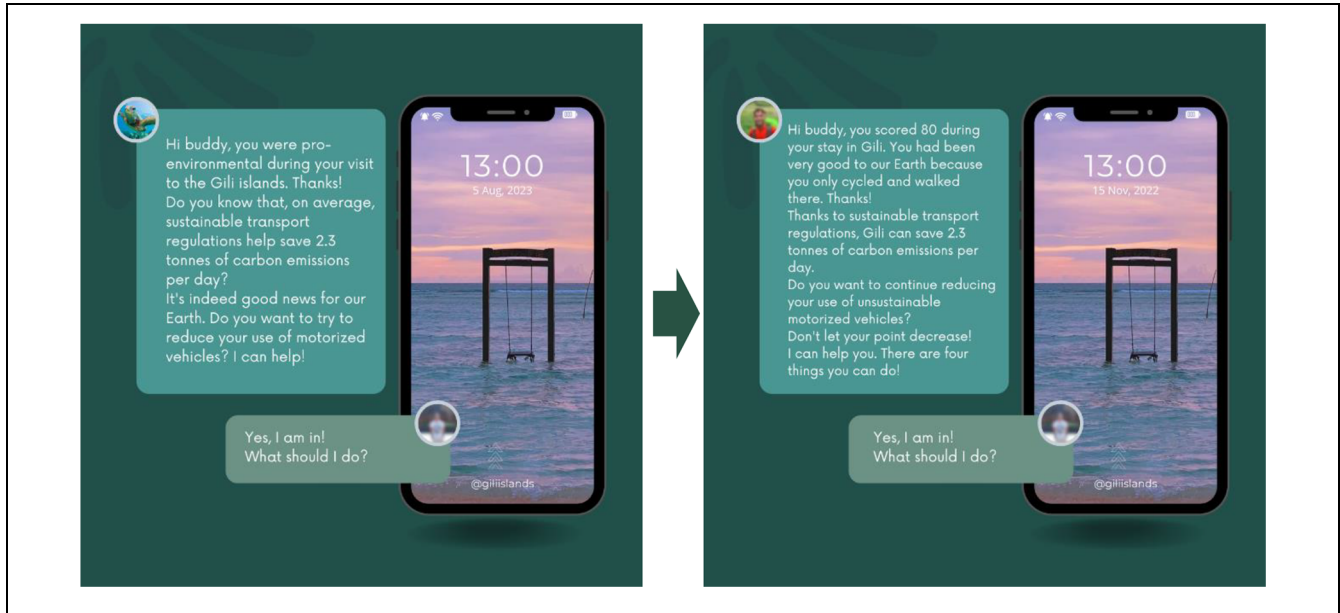


Figure 2. Development of prototypes for the proposed chatbot technology.
 Note. (Left: presented during data collection; Right: presented after data collection).

relieved their tourism memory (Y. R. Kim et al., 2022) and imagined that they had the proposed technology already during their visit, following the mental simulation approach (Chi et al., 2021).

All interviews were conducted online between December 2022 and January 2023 via platforms such as Zoom and WhatsApp call. Each interview session lasted 1 hour on average. Participants were compensated for their time. All interviews began with the interviewer presenting the initial concept of the chatbot technology and its context (e.g., sustainable traveling practices in the Gili Islands, Lombok). The concept presentation featured prototypes, including pictures displaying the proposed technology and the atmosphere of the Gili Islands. Thereafter, each participant was asked to answer interview questions based on their expertise and background. Examples of interview questions are *“How would you improve the proposed technology design?”* and *“How would you react if this chatbot were to ask you to reduce your use of private vehicles?”*

Moreover, Service Design techniques such as co-creating journey maps and personas were applied when discussing the proposed technology (Stickdorn et al., 2018). Figure 2 shows adjustments to the conversation design and the chatbot’s persona following input from the participants. All interviews were audio-recorded, transcribed immediately and translated professionally, where necessary, from Indonesian/Javanese to English. All transcripts containing languages other than English were translated into English by the main author, a native Indonesian and Javanese.

The themes that emerged from the interviews were categorized and developed into two frameworks that would later be confirmed with the participants through the FGDs. The FGD was chosen as the method for validating the frameworks since it can enable the participants to interact and generate more fully articulated accounts of a phenomenon (Wilkinson, 1998). All interview participants were invited to join the FGDs to validate the developed frameworks (Zhang et al., 2021). Two FGDs (group of six) took place in January 2023 sequentially (Nyumba et al., 2018). The FGD facilitator started the FGDs by reminding the participants of the proposed technology and sharing the developed frameworks and any changes made to the initial concept. The FGDs were conducted in Indonesian, audio-recorded, transcribed, and translated professionally.

Once all data had been gathered, the coding process continued to add evidence from the excerpts to the developed frameworks. Once the frameworks and designs had been completed, participant observation was conducted to further triangulate the data, allowing the researchers to see the proposed technological concept through the experience of the potential users in the field (Zhu et al., 2020). The first author stayed in the Gili Islands for 1 month in March 2023 to engage in conversations with relevant stakeholders such as tourists, hotel owners and locals. For instance, when engaging with tourists, the author explained the concept of the proposed technology and shared the research progress with them. Their feedback was noted and used for further consideration when refining the developed theoretical model (see Figure 1).

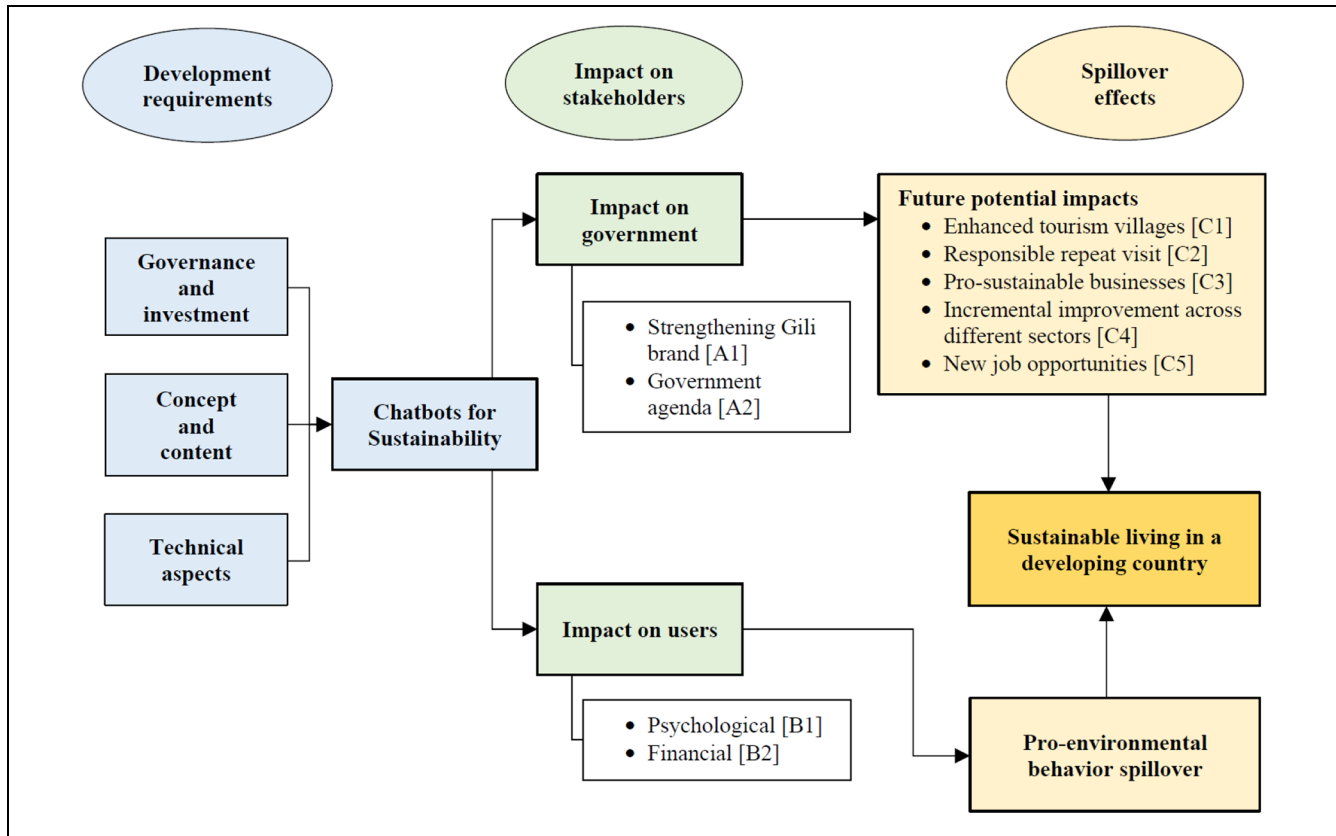


Figure 3. Overview of the implications of the development of the proposed chatbot technology.

Overall, the data analysis followed four coding steps (Corbin & Strauss, 1990; Wiesche et al., 2017): open coding (i.e., breaking down data analytically), axial coding (i.e., developing categories), selective coding (i.e., unifying categories around a core category) and theoretical coding (i.e., providing additional details about relationships between core categories). An independent researcher was trained to assist with two initial coding processes, while two other authors helped with the selective and theoretical coding stages. Disagreements were resolved through discussion. Constant comparisons between categories and concepts were performed during the analysis to ensure that the procedures were rigorous and systematic (Corbin & Strauss, 1990). The final version of the framework was communicated to the participants from the second FGD for their feedback and confirmation to ensure that the framework accurately reflects the discussion results (Muharam et al., 2024). A literature review was conducted to assist in explaining the derived concepts (Saunders et al., 2019).

Findings and Discussion

This section will be divided into two sub-sections to address two formulated research objectives. Section “Overview of the Implications of the Development of the

Proposed Chatbot Technology” will conceptualize the chatbot and discuss its potential implications. Then, Section “A Theoretical Model of Chatbots for PEB Spillover” will identify enablers of and hindrances to PEB spillover effects achieved through the conceptualized chatbot.

Section “Overview of the Implications of the Development of the Proposed Chatbot Technology” discusses three major categories, showing the steps resulting from developing the proposed chatbot. These include (1) development requirements, (2) impact on stakeholders, and (3) spillover effects. Figure 3 shows how introducing a technological concept herein referred to as “Chatbots for Sustainability” could result in the creation of sustainable living in a developing country.

Overview of the Implications of the Development of the Proposed Chatbot Technology

To begin with, selecting the Gili Islands as the context for the technology was deemed appropriate for several reasons. The Gili Islands are vulnerable areas that require particular care due to the impact of climate change on small islands (Hiwasaki et al., 2015). A local government official reported, “there’s [coastal erosion]. This is quite severe [as] the sea water level has increased” (D5,

interview). Furthermore, an environmental activist based in Gili Trawangan suggested, “*since Trawangan is located in the middle of the ocean, where will the pollution go? It stays here. The weather is already hot. Obviously, Trawangan is far from the mainland. Anything we need takes time*” (D7, interview). With such concerns in mind, implementing the customary law prohibiting the use of motorized vehicles is how the locals mitigate the natural consequences that arise from tourism development on and around the islands.

While the aforementioned internal factor justifies the development of the proposed technology, an external factor relating to the government further supports this proposition. A Ministry of Tourism and Creative Economy, Indonesia staff member proposed, “*Gili can be a destination that will support Mandalika. The tourists who have been to Mandalika can go straight to Gili for a vacation*” (D8, interview). Mandalika is part of the ten “New Bali” destinations, that received heavy investment for the establishment of an international circuit in 2022 to host MotoGP, the world-class motorcycle racing event (Taupikurrahman & Suwandana, 2022). She stated:

... we have a circuit in Mandalika, West Nusa Tenggara. ... so, it is how we can keep developing [the tourism industry there] moving forward because ... the contract [with MotoGP] is [for] multiple years. So, we [the Ministry of Tourism] and the regional government must think about what will happen once the contract is not extended” (D8, interview).

Such a line of thinking reflects the principles behind sustainable tourism development, which concern the long-term impacts of any tourism development project (United Nations Environment Program & World Tourism Organization, 2005). This is to avoid having abandoned large infrastructures, which often happens to mega-event hosts (Curi et al., 2011). The above proposition will leverage the concept of the tourism spillover effects, promoting local prosperity and employment opportunities (Y. R. Kim et al., 2021).

Development Requirements. When proposing a new technology, one needs to answer several critical questions to ensure that the idea meets all the requirements for developing the technology. The data analysis revealed three sub-categories that capture relevant concerns related to the development of the proposed Chatbots for Sustainability.

Governance and Investment. Behind the development of an innovation, questions over which organization is considered the best to tackle the challenge at hand become crucial to answer (Stickdorn et al., 2018). Many innovations fail to become sustainable because of the difficulties in finding a suitable business model that would enable

them to survive (Baldassarre et al., 2017). The proposed chatbot technology is no different.

Crucially, interviewees recognized that the proposed chatbot technology would be better implemented by the government, not the private sector. For the private sector to be able to implement technology to tackle such sustainability issues in Indonesia, one participant explained that the government would need to push public policy and regulations in a way that could make doing so attractive. He stated, “*[i]n terms of business models, such endeavors [in Indonesia] currently are not attractive enough for investors*” (T1, interview). Thus, the central government is deemed suitable to lead the investment in developing and implementing this technology. The investment will signal the positive direction that the government is taking toward sustainability and provide a template for the private sector to follow suit.

Regarding the target audience, the central government could start with domestic tourists rather than international ones. International tourists “*are outside of the Indonesian jurisdiction*” (S1, interview), making it difficult for the government to measure the program’s impact. However, to get the central government to invest in such innovation, Participant D8 suggested that it is necessary to work with the regional government that is responsible for the development of the Gili Islands first. This is to ensure that the proposed innovation represents the needs of the locals, reflecting a bottom-up development approach (Sakata & Prideaux, 2013). However, once the locals are engaged in the discussion, there is an increased likelihood of the project being received with skepticism. As such, conducting a preliminary feasibility study for an innovative government project can help provide more substantial justification for the proposed project (Yoon, 2018).

The conclusion that the central government is the ideal stakeholder in developing this concept corresponds to the current strategy of the Indonesian tourism development plan (Kemenparekraf, 2022). One participant stated, “*... the Ministry of Tourism’s programs already go in that direction [sustainability]. In 2018, [the Gili Islands] were hit by an earthquake. So, the big picture includes resilience, sustainability, and inclusivity*” (D5, interview). Past research has suggested that technological innovation should be aligned with the respective government’s agenda to increase the chance of success (X. Zhou et al., 2020). Another participant gave the assurance that “*if the political willingness of the government is strong, the program will work*” (S1, interview). This point is aligned with the theory of a strategic-relational political economy, which emphasizes that the government needs to show its political stance when directing a change (Bramwell, 2011). Once it has been confirmed that the central government is willing to invest in such a technological innovation, a further discussion involving relevant organizations needs to take place to outline the key objectives and measurement criteria of success behind the technological innovation.

Concept and Content. Since the proposed technology is in the form of a conversational agent, the starting point of the conversation itself becomes a critical point. According to the Service Design approach, this is part of the journey mapping stage (Stickdorn et al., 2018). The data reveal two scenarios where the chatbot can start conversing with domestic tourists. The first one is during the visit, which can be implemented through collaboration with hotel owners on the islands. A business developer at a chatbot startup suggested:

You can put the QR code at the hotel. So, once they [tourists] enter the room, they are healed and ... relaxed. They would start exploring, *“what am I going to do next? Oh, there’s a [QR] code in my room”* (T2, interview).

Another scenario is to start the interaction at the “pre-visit” stage through collaboration with tour operators. They can start sharing access to the chatbot before the tourists embark on the trip (Carvalho & Ivanov, 2023). The chatbot can be designed as a travel companion that can provide information regarding the sustainability facts of the Gili Islands.

So, before the flight, ... we talked about [my] plans for the vacation. ... We set the chatbot to trigger first. So, it isn’t just a response, it is pro-active in asking questions. So, we set her up in the morning to broadcast to my number. *“Morning Andrew [pseudonym], how are you? Your flight is today. Don’t leave anything behind. Let [me] know when you’re at the airport.”* So, once I arrived at the airport, I chat, *“Beliza [the chatbot’s name], I’ve arrived at the airport. I’m going to have a flight soon.”* Later, Beliza can answer, *“well, have a safe flight. ... Beliza just wants to make sure you have a safe flight.”* So, keep it very engaging like that. (T2, interview).

In this way, the proposed technology could cover all the traveling stages: before, during, and after the trip. The fact that the users are already familiar with the technology before they leave the islands could help maintain their engagement post-trip if the chatbot is designed to promote pro-environmental behavior spillover.

Concerning the content of the proposed chatbot technology, perceived usefulness is one of the most crucial factors driving technology adoption among users (Davis, 1989). People would evaluate whether using the technology could help them meet their goals. Therefore, designing technological solutions for sustainability reasons can be challenging since there is less immediate effect felt by the technology users (Naderi & Strutton, 2015). To solve this issue, a startup CEO suggested the following solution:

Digital products can be widely adopted, like, Gojek, Grab, and Tokopedia, ... with a promo. ... If you ride this bike,

you will save a lot, you can get a voucher, ..., at restaurant A, B, and C. It needs to be presented like that, especially in the early stages, to increase awareness... [People will ask] *“What’s in it for me?”* (T3, interview).

However, maintaining user engagement is another challenge that will have to be faced after the initial adoption stage. The information the proposed chatbot provides should be helpful and relevant to users. It should be able to either enhance their experience on the Gili Islands or effectively increase their environmental awareness. To ensure the messages are effective, an Indonesian working at a sustainable travel company in the UK suggested that content development should involve experts from multiple disciplines whose backgrounds would complement each other. She stated, *“... we, as practitioners in field A, cannot work alone. ... If you are trying to educate [people] through this chatbot program, there should be peer reviews from other experts ... [Such] projects ... must be people-centric.”* (S7, interview).

Since it is a conversational AI, most participants agreed that the language is key to users’ acceptance of the proposed chatbot technology. Participants D2 and D3 emphasized that it should be easy to understand, and any technical terms should be explained in order to reach the intended target users effectively. It would be even better if the conversation could be adjusted so that it follows the users’ characteristics, echoing the communication accommodation theory (Giles & Ogay, 2007). The startup CEO stressed the importance of “personalization,” stating *“take you as an example, since you are easy-going, maybe [go with] casual. Maybe with older people, [use] more polite language. ... In the future, chatbots will have to be personalized just like friends”* (T3, interview). This statement echoes the recommendation given in existing research on chatbots for pro-self-nudging, suggesting that personalized nudging will increase efficacy (Zhang et al., 2020).

Technical Aspects. With a chatbot, one must decide which platform it will be deployed on later so that users can access the service. Based on the collected data, WhatsApp was considered to be the most appropriate platform via which to deploy the chatbot as it is the most popular mobile messenger application worldwide, including in Indonesia (Ceci, 2022). The World Health Organization (WHO) also used the WhatsApp chatbot to orchestrate its global risk-communication outreach during the COVID-19 pandemic (Walwema, 2021).

Furthermore, participants with experience in chatbot development all agree that the execution of the project will start with designing the conversation to determine how the chatbot and its users will flow. An NLP expert stated, *“the conversation design is more intensive ... you have to design it yourself”* (T5, interview). This includes facts and figures that

are to be presented through the chatbot, which will be provided by the implementer (i.e., government organizations proposing the technology). However, this could change with the advancement of Generative Conversational AI such as ChatGPT. Another startup CEO suggested a more explorative route for developing the chatbot using the latest AI trend. He stated, “*now creating a chatbot is easier ... Your task is to create a good standard operating procedure for the chatbot. Consider ChatGPT as someone you can teach*” (T4, interview). This statement can be considered when seeking to solve the technical challenges that emerge when one aspires to create a sustainable travel companion, as pointed out in a recent discussion on ChatGPT for tourism (Carvalho & Ivanov, 2023).

Chatbots for Sustainability. The following discussion on the potential impact of the proposed chatbot technology will be divided into two sub-sections. The first sub-section presents the potential immediate impact on the government (as the assumed investor) and the broader impacts (i.e., the spillover effects). More specifically, the spillover effects encompass aspects of the country’s development that are expected to emerge from implementing the proposed chatbot technology. How technology would eventually affect changes, in the long run, is naturally debatable. Therefore, the discussion on the potential spillover effects of the chatbot implementation is expected to spark debates that could result in more fruitful academic discussions and scientific endeavors.

The areas covered in the second sub-section include the potential immediate impact on the users—Indonesian tourists—and the spillover effects. Since the proposed chatbot is a behavior change intervention medium, subsequent behavior that is influenced by its use, that is, the PEB spillover, will be the focus of the second theoretical discussion in this study.

Impact on the Government. Successful technological innovations bring positive public perception for innovators (Ma et al., 2017). Taking the Gili Islands as the study context, the islands and their respective tourism stakeholders will be the first to benefit from the creation of the proposed chatbot. A local tourism official suggested:

This chatbot is to strengthen Gili’s branding as an environmentally friendly destination. ...if we look back at the market for tourists who come to Gili, it is dominated by foreign guests. ...one of the government’s [intentions] ... [is to] encourage domestic tourists ... [to] stay longer” (D5, FGD).

Furthermore, since a chatbot can convey messages, it can be used as a platform by the government to communicate its national agenda to the public. For instance, good communication of the government’s commitment to the

sustainability agenda to the public will ultimately improve public awareness of sustainability (Font & McCabe, 2017). For instance, one participant suggested:

... according to the blueprint of the Indonesian government, Indonesia wants to be the biggest electric battery producer in the world. ... When you return from Gili, you don’t have to take public transport, so it’s like, “*Are you interested in being greener?*” The answer, for example, is “yes.” [They are] later given the choice of moving to public transportation, using an electric vehicle, or cycling. We nudge them into choosing electric [vehicles] (S1, interview).

In such a scenario, the government can campaign and advocate for the benefits of switching to greener, alternative transport options by highlighting its positive environmental impact.

Another positive outcome of this innovation is that developing countries like Indonesia may demonstrate their ability to bring groundbreaking technological innovations. A researcher working on sustainability in Indonesia stated, “*the Indonesian people are ... followers of trends*” (S2, interview). Developed countries often set trends in technological innovations (Krugman, 1979). Innovations initiated in developing countries like Indonesia will help increase global competitiveness (Aubert, 2005). The Ministry of Tourism staff member further commented that “*[this innovation] can help raise the name of Indonesia. So, it’s like people are starting to pay more and more attention; we are the first to get this concept*” (D8, interview).

Spillover Effects from the Impact on the Government. Relevant to previous discussions, one of the programs sponsored by the Indonesian government is the development of “Tourism Villages.” These are tourist destinations built on community-based tourism and ecotourism principles where the locals will benefit from the tourism activity (Nugroho et al., 2018). These tourism villages can condition the tourists to be more environmentally mindful during their visit. The government could leverage these 4,668 villages (Kemenparekraf, 2023) to serve as places where transformative tourism experiences could occur by facilitating the meaning-making processes among tourists through their interactions with the chatbot technology.

However, an enhanced tourist experience might cause a boomerang effect on the sustainability agenda as tourists may repeat their visit. Tourists finding the place they visited to be memorable would come again, and their trip would consequently emit carbon. Through field observation, an environmental activist from France based in Gili suggested that although the tourists’ traveling behavior on the islands could be considered environmentally friendly, they have traveled a distance with considerable carbon emissions along the trip. A potential solution is to

introduce responsible visits through schemes such as carbon offsetting. The destination management officer commented on this idea:

As long as there are rules and regulations that we can formalize, I don't think it's a problem and [as long as it] won't burden tourists. ... We [will] support the innovation [with respect to] the carbon [offsetting] ... as we will know that people coming here are already emission-free (D4, interview).

The proposed chatbot can act as an agent to inform people of the impacts of the carbon offsetting program to create transparency (Babakhani et al., 2017).

The proposed chatbot technology can also have an impact on businesses such as hotels, restaurants, and tour agencies. They can be encouraged to support the sustainability agenda by providing incentives to pro-environmental tourists, such as coupons (Peng & Lee, 2019). The startup CEO suggested, *"it would be good if this government initiative is supported by NGOs, ... local business people ... restaurants A, B, C, and [customers can] get a discount. So, the win-win solution [is] in the [creation of an] ecosystem [for promoting sustainability]"* (T3, interview). This approach would create an ecosystem of pro-sustainable businesses.

Once key stakeholders have shown their support for the sustainability agenda implemented by the government, it will trigger incremental improvements across different sectors. For instance, education will be the ultimate sector that benefits from the change introduced here. A psychology researcher commented:

Let's say [the tourists on the Gili Islands are] teenagers and young adults ... when they have matured as individuals, they already have ... a positive attitude [toward the environment]. Project this to, say, 10 years, 20 years into the future ... from the side of human resources investment, [it] is extraordinary (S4, interview).

In this way, technological innovation ultimately invests in the nation's "character building." It is necessary to do so because unlike European countries such as the UK, Indonesia has not incorporated sustainability into the curriculum. One participant stated, *"education related to environmental awareness needs to be instilled from an early age, ... in the curriculum in kindergartens and elementary schools. ... we [Indonesia] are [currently] not there [yet]"* (S3, interview). This statement was echoed by a sustainability activist from the UK when she reflected on her experience growing up in the UK. She stated:

I think [education on sustainability] is actually part of the curriculum. Not a very big part, but in certain lessons, we

learn about fossil fuels. We learn the words "finite" and "infinite" ... at the age of probably 8 years old (S6, interview).

Bearing this in mind, one can argue that implementing this chatbot technology will shed light on the sustainability awareness of Indonesian society, which, in the future, can drive the younger generation to be more environmentally mindful.

New job opportunities, such as those in the electric vehicles sector, will also be unlocked. One participant stated:

When there is a conversion workshop [for electric vehicles]; there is a derivative industry. There is a homemade battery industry run by electronics geeks in the city ... if you buy it, it's expensive—10 million [Indonesian Rupiah]. If assembled by them, it's only 3 million [Indonesian Rupiah] (S1, interview).

This shows how the spillover effects of creating a technology can alter the economy's direction in one country and the behavior of its people (Azzuhri et al., 2018).

Impact on the Users and PEB Spillover Effects. The data analysis revealed two identifiable patterns when interviewees were asked how users would respond to the nudges provided by the chatbot. The first relates to how the messages presented to them are processed psychologically, and the second is the financial implications of changing their behavior. In essence, these two aspects (i.e., psychological and financial) of the human decision-making process reflect the findings in the past literature on the drivers behind the adoption of PEB (Bolderdijk et al., 2013).

Firstly, the analysis revealed that users would realize they had a new environmental identity after learning from the chatbot that they had been pro-environmental during their visit (Van der Werff et al., 2014). One participant suggested that *"... the experience in Gili can be an extraordinary trigger for them to become [environmentally] aware"* (S3, interview). Past research has shown that initiating behavior change by increasing one's sustainability awareness will result in a transition to more sustainable behavior compared to if it is initiated only by financial appeals (D. Schwartz et al., 2015). Moreover, the nudges given to the users would also provide them with a signal concerning the direction in which society is moving. If more people are aware of sustainability issues and embrace PEB, then the perceived social norms will serve as a mechanism to sustain the change beyond using the chatbot itself (Culiberg & Elgaaiied-Gambier, 2015).

However, aligned with the concept of perceived behavior control in the Theory of Planned Behavior, if the users feel that they do not have the resources to follow

the nudges (Ajzen, 1991), then they will feel frustrated and guilty. One participant expressed this when asked about his reaction to the hypothetical continual reminder sent by the chatbot:

I am [from] a community with [limited] human resources and economic power. What is your deal [referring to the chatbot]? I'm uncomfortable. ... Why are you reminding me? Do you have any [tangible benefits] for me? On the other hand, I need [my private motorcycle]. If [the public transportation] is expensive, what should I ride? If you don't give me money, why do you ban me? (D2, interview).

These feelings would stop the PEB spillover from occurring, and people would continue the old behavior despite the increase in their environmental awareness. Indeed, the above statement from the tourist's perspective aligns with several important factors that have been found to predict people's willingness to use environmentally friendly transportation, such as perceived efficiency from time and money points of view (Buys & Miller, 2011) and the perceived tangible financial support from the government, including subsidies (Sierzechula, 2014).

The participants further mentioned several factors that they believed would significantly facilitate the PEB spillover. One participant suggested that *"apart from socio-demographics ... [such as] income and education, location also has an effect"* (S3, interview). For instance, taking public transportation in an area where it is unreliable would be risky. One participant suggested, *"this will be difficult depending on where you are ... we have a mindset [aiming for] efficiency ... [and] people are calculating the risks [of taking alternative transportation]"* (S2, interview). Such complexity leads to the last factor suggested by the participants, that government support will influence societal change. If the transport and infrastructure quality is left unimproved, people will not embrace PEB spillover. One participant stated that *"[the change in our behavior will] depend on the government. If that's what's implemented, we'll comply"* (D3, interview). Therefore, for a PEB spillover to occur, government policy and regulations need to be aligned with the efforts of the people.

Sustainable Living in a Developing Country. Despite evident challenges surrounding the implementation of the proposed chatbot technology, it provides an opportunity to create a sustainable living environment in a developing country like Indonesia. Tackling sustainability challenges has been widely recognized as an expensive endeavor requiring tremendous financial resources. Research has shown that economically developed countries are more ready for sustainability transitions (Wieczorek, 2018).

With nature-based destinations encouraging tourists to behave pro-environmentally in developing countries such

as Indonesia (Scheyvens, 1999), the invention of a technology that can facilitate self-reflection among tourists could enable transformative tourism experiences. With incremental improvement taking place across many other sectors, the ecosystem that can support sustainable living in the country will eventually be formed. One activist provided the following comment that is aligned with Sustainable Development Goal target 17.7 from the 17 goals of the United Nations 2030 Agenda for Sustainable Development on the importance of the transfer of environmentally sound technology to developing countries:

We have to start investing in sustainable technologies for developing nations and pulling them off the ground. Then that's the kind of sustainable way for everybody ... that comes from needing better funding but also better education and school and increasing awareness in general (S6, interview).

The above discussion concerning the proposed technological concept and its implications highlights how chatbots can potentially be promoted to facilitate PEB spillover among tourists visiting nature-based destinations. The following sub-section will outline a theoretical model in order to highlight important factors that predict the successful implementation of the proposed chatbot technology.

A Theoretical Model of Chatbots for PEB Spillover

Based on the findings, four categories of factors predict how the proposed chatbot technology will bring positive changes to the users' traveling behavior. The first two categories predict the use of the proposed chatbot: (a) technology-related heuristic assessment and (b) user-related heuristic assessment. The other two predict the PEB spillover in the use of environmentally friendly transport: (c) transport-related heuristic assessment, and (d) socio-demographic factors.

The first category of constructs, technology-related heuristic assessment, concerns how people evaluate the technology itself, including how it performs and what it can offer. Four constructs are identified under this category. As stated in the literature review, the first two constructs still reflect factors in existing technology acceptance theories such as TAM, UTAUT and AIDUA, namely Performance Expectancy and Effort Expectancy (Venkatesh et al., 2003). These constructs are often the strongest predictors because users constantly evaluate them before adopting any technology.

The first construct, "Performance Expectancy," refers to the perceived benefits that the technology can bring users through its performance (Venkatesh et al., 2003). However, in this study context, users expect to see

something more tangible from using the technology, such as a reward. This is because contributing to the environment is perceivably unnoticeable and insignificant, therefore discouraging people from making extra effort to do so (Viana et al., 2022). This highlights the challenges that technology innovators face in making the issues of tackling sustainability interesting for people (Gholami et al., 2016). Designing technological solutions for sustainability problems in a way that highlights the benefits that users can gain is, therefore, crucial.

The second construct, “Effort Expectancy,” includes the extent to which the users feel that using the technology is easy (Venkatesh et al., 2003). Previous literature on chatbot adoption found that it is a strong predictor of usage intention alongside “Performance Expectancy” (e.g., Hanji et al., 2023; Pillai & Sivathanu, 2020; Zhang et al., 2022). In this study context, the user evaluation of the given effort centers around specific aspects of the proposed chatbot technology, such as language and access. The third construct is “Personalization,” referring to the perceived ability of the proposed chatbot technology to personalize its communication style and services to appeal to the users. Aligned with research on chatbots as a behavior change medium, personalization is the next step that innovators can explore to effectively deliver intervention messages (Zhang et al., 2020). This correlates with the findings of Zhang et al. (2022), which demonstrate that “Personalization” has a positive effect on the intention to continue chatbot usage.

The last construct in the technology-related heuristic assessment is “Credibility.” Because the chatbot technology proposed in this study is expected to be part of the government program designed to help achieve its sustainability agenda, “Credibility” herein refers to how trustworthy the technology is perceived to be. Especially in today’s era of increased government surveillance, data phishing, and online scamming, people are keen to know with whom they are interacting online (Chen et al., 2017), especially on a private messaging platform such as WhatsApp. Past research demonstrated that “Credibility” is an important factor in people’s decision to trust information shared online (e.g., Fogg et al., 2001).

The second category of constructs, user-related heuristic assessment, can be defined as how the users perceive their situation when interacting with the proposed technology. The first construct is “Privacy Control,” which is associated with how users would demand their ability to give consent when they want to start the interaction with the proposed chatbot, how frequently they would receive intervention messages, and how their data would be managed (Chen et al., 2017). People expect to be able to exercise their control over technology despite wishing to receive more personalized services, which encapsulates the personalization-privacy paradox (Awad & Krishnan, 2006).

The second construct is “Habit,” which refers to how frequently and habitually the users use messaging applications, such as WhatsApp, as their communication medium will influence their use of the proposed technology. Previous research on chatbots for tourism found that “Habit” is an important factor in people’s decision to adopt the technology (Melián-González et al., 2019; Zhang et al., 2022). The last construct found in this category is “Timing,” which refers to how the users perceive the timing of the messages sent by the proposed chatbot to align with their availability. Purohit and Holzer (2019) indicated that “Timing” is a crucial factor in the digital nudging context despite being an under-researched factor in the context of technology acceptance theories.

These two categories are expected to lead to the adoption and continued use of the proposed chatbot technology. Nevertheless, because using the technology will lead the users to follow the nudges provided by the chatbot, it is also essential to look at the predictors of the target behavior, which is the PEB spillover in the form of the use of environmentally friendly transport (Figure 4).

The first category of predictors of spillover behavior is transport-related heuristic assessment, which consists of four constructs. The first construct, “Accessibility,” refers to people’s assessment of the accessibility of environmentally friendly transport in their region. Previous research on sustainable transport and nudging noted that “Accessibility” is an important factor to consider when encouraging people to change their mobility patterns (e.g., Anagnostopoulou et al., 2020). For instance, the uneven infrastructure development in countries such as Indonesia can make some environmentally friendly transport options inaccessible (Aswicahyono & Friawan, 2008). This is related to the second construct, “Efficiency,” which refers to people’s evaluation of the level of efficiency of taking alternative transport options based on three criteria: cost, effort and time. This factor is aligned with the suggestion from previous literature that studied people’s determinants in their decision to travel (Ahn & Park, 2022; Buys & Miller, 2011). Furthermore, Stephenson et al. (2018) found that efficiency is also a key reason that people take unsustainable transport options despite having high environmental awareness.

The final construct, Government Support, is associated with the extent to which people feel that the government is supporting the citizen’s adoption of environmentally friendly transport options. For instance, if people feel that there is no improvement to the quality of the public transportation system after switching (Nykqvist & Whitmarsh, 2008), they might return to the old behavior (of using private vehicles) despite an increase in their environmental awareness. Other than quality improvement, government support can be financial, such as subsidies or discounts, as suggested by past research (Sierzechula, 2014).

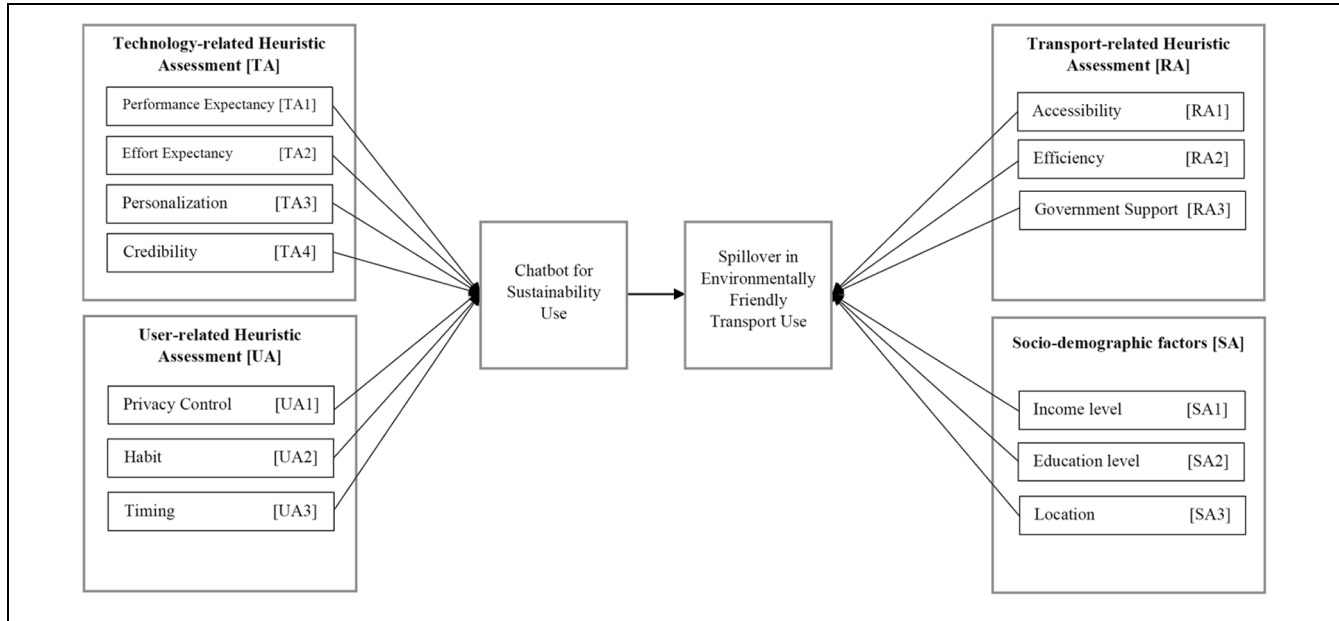


Figure 4. A theoretical model of chatbots for PEB spillover.

The second category of spillover behavior predictors includes three variables influencing how people perceive the use of environmentally friendly transport options based on their socio-demographic characteristics. As the first variable, “Income” significantly predicts how people would switch to environmentally friendly transport options given that transitioning to more sustainable behavior will alter their economic expenses (Papagiannakis et al., 2018). Those with a higher income level would have more options than the lower-income group. However, the higher the people’s income, the more they value their time. This tendency can drive people to use even more unsustainable transport options, such as private jets, for higher efficiency (Barros & Wilk, 2021). The next variable is “Educational Level.” Past research has shown that people with higher education levels tend to be more aware of environmental issues (Sudarmadi et al., 2001). Thus, they are expected to be more open to behaving more sustainably after receiving the nudges. The last variable is “Location” because access, infrastructure and transport quality will differ depending on where people live, and all three will influence their evaluation of the potential for behavior change.

Conclusion and Implications

With the rapid advancement of conversational AI, this study aims to conceptualize a chatbot technology that can facilitate PEB spillover among tourists who have been to the Gili Islands, a nature-based destination. To achieve this aim, this study conducts exploratory qualitative research employing the Service Design approach and

Grounded Theory. This study enriches the literature on nudge theory by exploring new ways of nudging through conversational AI chatbots (Zhang et al., 2020). Pro-social nudging through chatbots is an understudied phenomenon. The selection of continued use of environmentally friendly transport among tourists as the target behavior of the pro-social nudging further demonstrates this study’s theoretical contribution since research on the PEB spillover emerging from the tourism context is still in its infancy (Souza-Neto et al., 2022).

Benefitting from multidisciplinary perspectives and diverse expertise of participants (i.e., technology development, social sustainability, and destination management), this study makes significant theoretical contributions by developing two frameworks. First, the general framework providing an overview of how the proposed chatbot technology can be developed will address the call from Gholami et al. (2016) and Tussyadiah (2020). Tackling sustainability issues through AI-based innovations remains a challenge, specifically in finding the right stakeholders to explore this type of endeavor (Baldassarre et al., 2017). Understanding that the government is the ideal stakeholder to initiate and support this innovative solution is crucial to progressing the discussion on how AI can help tackle sustainability issues. Furthermore, the expected future impacts and the spillover effects of the proposed technology will help the government better understand how a technological solution can potentially have positive implications for society (Azzuhri et al., 2018).

Second, this study proposes a theoretical model of the chatbots for PEB spillover to capture factors that are

predicted to influence the use of the proposed technology and the adoption of the suggested PEB spillover. Much of the past literature has struggled to explain why modifying people's green behavior through technology is complicated (e.g., Gholami et al., 2016). Specifically, the transport-related heuristic assessment explains how pro-social nudging would always entail people evaluating their perceived ability to follow the nudges. Despite an increase in their environmental awareness, it could be that the current level of perceived behavior control is still too challenging to overcome (Ajzen, 1991). Such considerations reflect the "rational man" hypothesis in traditional economics, which, according to Simon (1986), plays a significant role because rationality allows individuals to understand the essence of things in a logical manner.

However, the increased environmental awareness could also incrementally drive positive changes. According to the Norm Activation Model (NAM), the activation of individual moral norms affects pro-social behavior decision-making (S. Schwartz, 1977). Once they have started behaving pro-environmentally more often, it will create a new sense of self-identity, which can drive them even further into sticking with their pro-environmental behavior (Sparks & Shepherd, 1992). As such, although changes do not take place immediately, improvement can happen more organically as more people improve their environmental awareness through the behavior change intervention mechanism that has been put in place. In short, the proposed theoretical model expands the discussion on existing technology adoption theories such as TAM, UTAUT, and AIDUA.

Furthermore, this study presents a substantial number of practical implications for several key stakeholders. For policymakers, this study outlines the technological innovation that they can help to support the improvement of tourist experience in their regions. Businesses can also start exploring AI-based solutions to tackle sustainability issues with more confidence by ascertaining how their proposed solutions should be aligned with government support. For sustainability activists, access to AI technology such as ChatGPT will enable them to explore potential innovations that will make it possible to change people's behavior at scale. Lastly, governments of developing countries can leverage nature-based destinations that they have as sites where transformative tourism experiences occur by using technology such as chatbots.

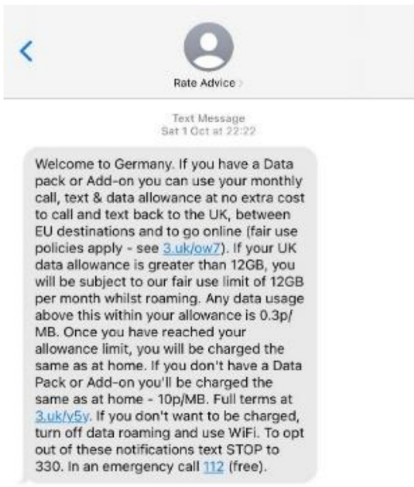
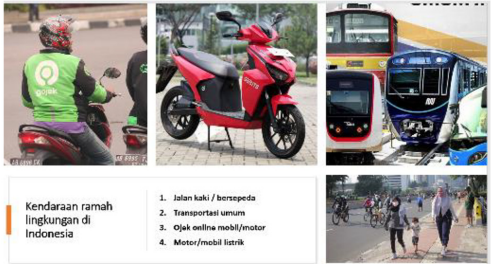
Generative AI such as ChatGPT can revolutionize how tour guiding works in a tourist destination (Dwivedi et al., 2023), helping tourists better understand their past positive experiences. A better reflection process will unleash PEB spillover potential (Kirillova et al., 2017). In the future, chatbots will be able to personalize their services (Carvalho & Ivanov, 2023), making learning processes more personalized (Zhang et al., 2020). The marginal positive contributions toward the conservation of the planet from a large number of tourists can then be aggregated to drive change thanks to the power of AI technology (Majid et al., 2023). As such, this study answers the calls from previous scholars regarding the need for research on AI technology in tourism that can contribute to environmental sustainability (M. J. Kim et al., 2023; Xiang et al., 2021).

Limitations and Future Research

Despite its contributions, this study has several limitations that future research can address. First, the Gili Islands, as the study context, represent a unique proposition in terms of sustainable transport regulations. Future research can replicate the concept of the present study and advance the investigation by leveraging other locations. Nudges can be adjusted to the particularities of destinations (e.g., environmental conditions, policy), such as by developing educational nudges to encourage resource conservation in destinations that are deprived of natural resources (e.g., water). Second, since Indonesia is still lagging in its efforts to address the sustainability agenda (Kanugrahan et al., 2022), developed countries may be able to explore more effectively the potential for having a chatbot to trigger a PEB spillover among tourists. With more resources, including funding for green businesses, developed countries may lead the innovation, and others will follow suit. Third, the theoretical model proposed in this study needs to be validated through quantitative research to support generalizability. A scenario-based experimental survey could be an option for a technology that is still in the conceptualization stage. Lastly, further research efforts should be undertaken to ensure that the proposed technology concept can be implemented. Longitudinal research using real human-chatbot interaction as an experiment can be among the options.

Appendices

Appendix 1. Service Design-Guided Presentation.

Phase	Detailed explanation	Supporting materials
Introduction to the study context	This picture was shown to give participants an idea of the sustainable travelling practices applied in the Gili Islands. It is helpful for informants who have never been to the Gili Islands. Moreover, the discussion on sustainability and its derivatives is still new in Indonesia. This picture can help informants understand technical definitions, such as zero emissions and active travel. This picture was shown to participants so they know which platform the chatbot will most likely be deployed on. WhatsApp is considered the most appropriate since it is the most used messaging platform globally. Targeting an app people are already familiar with will increase the sense of getting personalized services. This picture was presented to show participants how the chatbot can automatically detect the tourists when they have left the Gili Islands. This is a picture of Rate Advice SMS, a message that people automatically receive when landing in a new country abroad. However, after discussing it at length, the concept is considered not feasible as there is a challenge in collecting the tourists' contact information. Then, the discussion revolves around how to solve this issue, discovering two different routes when starting the conversation—pre-visit and during the visit.	  
Presentation at the start of the FGD session	This collation of pictures shows environmentally friendly transport options available in Indonesia. There are four options: online ride-hailing services (represented by the man in the green jacket), electric vehicles (represented by the red motorcycle), public transport (represented by a picture of trains and buses), and active travel (represented by the image containing people walking).	 Kendaraan ramah lingkungan di Indonesia 1. Jalan kaki / bersepeda 2. Transportasi umum 3. Ojek online mobil/motor 4. Motor/mobil listrik

Appendix 2. Interview Guide Questions.

Category	Examples
General opening questions	- What do you think of the proposed technology concept? - Have you ever visited the Gili Islands? If yes, when do you think the conversation between the chatbot and the tourists should start?
Questions asked to participants in the “Technology and Process Development” group	- Imagine that this technology exists; what should the conversation be like? - How would you improve the proposed technology design? - Do you think that creating such technological innovation is feasible? - What are the aspects that I should take into consideration when proposing this concept? - If I want to start developing this technology, what should I do first? - What are the challenges for me if I want to continue developing this technology? - Since we use artificial intelligence to increase personalization, do you think data privacy issues will be relevant to the Indonesian people?
Questions asked to participants in the “Social Change and Sustainability Preparedness” group	- Do you think Indonesians are aware of sustainability issues and ready to become more environmentally mindful? - Who should be the investor of the proposed technology? If the government, which level? Central or regional? How should I start approaching them? - When discussing environmentally friendly transportation in Indonesia, how difficult is it for people to reduce their use of private vehicles? - What should we do in order to facilitate behavior change among technology users better so they can use environmentally friendly transport options more often?
Questions asked to participants in the “Destination Management and Experience” group	- What are the environmental and social challenges that the Gili Islands are currently facing? - Do you think people coming to the Gili Islands already have high environmental awareness levels? What are the demographics like? Maybe you can explain how domestic and international tourists behave during their visit? - Which positive experiences in Gili should I capture and present in this technology? - Who will benefit from this technological innovation? - How would you react if this chatbot were to ask you to reduce your use of private vehicles? - If I managed to develop this technology, what should I do to increase your willingness to use it and for you to follow its suggested actions?

Appendix 3. Codebook.

Level 1 Code	Level 2 Code	Level 3 Code	Code Name	Definition	Sample Quote
DEV			Development requirements	Any fundamental aspects that need to be considered before embarking on the development process of the technology	“There’s [coastal erosion]. This is quite severe [as] the sea water level has increased” (D5, interview) “Since Trawangan is located in the middle of the ocean, where will the pollution go? It stays here. The weather is already hot. Obviously, Trawangan is far from the mainland. Anything we need takes time” (D7, interview) “... we have a circuit in Mandalika, West Nusa Tenggara. ... so it is how we can keep developing [the tourism] moving forward because as I remember, the contract [with MotoGP] is [for] multiple years. So, we [Ministry of Tourism] and the regional government must think about what will happen once the contract is not extended” (D8, interview)
	OI		Governance	The identification of appropriate organization that should be responsible for promoting the innovation	“[I]n terms of business models, such endeavors [in Indonesia] currently are not attractive enough for investors” (T1, interview) “They are outside of the Indonesian jurisdiction” (S1, interview) “... the Ministry of Tourism’s programs already go in that direction [sustainability]. In 2018, we were hit by an earthquake. So, the big picture includes resilience, sustainability, and inclusivity” (D5, interview)

(continued)

Appendix 3. (continued)

Level 1 Code	Level 2 Code	Level 3 Code	Code Name	Definition	Sample Quote
	OI		Investment	The potential allocation of any resources needed to invent the technology	"If the political willingness of the government is strong, the program will work" (S1, interview)
	CC		Concept	The general concept of innovation, including the flow of human-technology interaction	"You can put the QR code at the hotel. So, once they [tourists] enter the room, they are healed and ... relaxed. They would start exploring, <i>'what am I going to do next? Oh, there's a [QR] code in my room'</i> " (T2, interview). "So, before the flight, for example, we talked about [my] plans during the vacation. ... We set the chatbot to trigger first. So, it isn't just a response, it is pro-active in asking questions. So we set him up in the morning to broadcast to my number. <i>'Morning Andrew [pseudonym], how are you? Your flight is today. Don't leave anything behind. Let [me] know when you're already at the airport.'</i> So, once I arrived at the airport, I chat, <i>'Beliza [the chatbot's name], I've arrived at the airport. I'm going to have a flight soon.'</i> Later, Beliza can answer, <i>'well, have a safe flight. ... Beliza just wants to make sure you have a safe flight.'</i> So, keep it very engaging like that" (T2, interview)
	CC		Content	The specific topics, offers and messages delivered by the technology	"Digital products can be widely adopted, like, for example, Gojek, Grab, and Tokopedia, ... with a promo. So, for example, if you ride this bike, you will save a lot, you can get a voucher; ..., at restaurant A, B, and C. It needs to be presented like that especially in the early stages to increase awareness... [People will ask] <i>'What's in it for me?'</i> " (T3, interview) "Take you as an example, since you are easy-going, maybe casual. Maybe with older people, [use] more polite language. ... So, I think in the future chatbots will have to be personalized just like friends" (T3, interview).
	TA		Technical aspects	Information related to the basic technicalities to support the development of the technology	"The conversation design is more intensive ... you have to design it yourself" (T5, interview) "Now creating chatbot is easier ... Now your task is to create a good standard operating procedure for the chatbot. Consider ChatGPT as someone you can teach" (T4, interview)
IMP			Impact on stakeholders	Immediate potential impacts brought by the proposed innovation on key stakeholders	
A			Impact on government	The government is identified as the key stakeholder since it is the assumed ideal investor in this study context	
		A1	Strengthening Gili brand	Information related to how the technological innovation potentially helps the tourism industry in the Gili Islands	"This chatbot is to strengthen Gili's branding as an environmentally friendly destination. ... if we look back at the market for tourists who come to Gili, it is dominated by foreign guests. ... one of the government's [intentions] ... [is to] encourage domestic tourists ... [to] stay longer" (D5, FGD)
		A2	Government agenda	Information related to how the chatbot would help the government to socialize its national agenda	"... according to the blueprint of the Indonesian government, Indonesia wants to be the biggest electric battery producer in the world. ... When you return from Gili, you don't have to take public transport, so it's like, <i>'Are you interested in being greener?'</i> The answer, for example, is <i>'yes.'</i> [They are] later given the choice of moving to public transportation, using an electric vehicle, or cycling. We nudge them into choosing electric [vehicles]" (S1, interview)

(continued)

Appendix 3. (continued)

Level 1 Code	Level 2 Code	Level 3 Code	Code Name	Definition	Sample Quote
B			Impact on user	The user in this study context refers to the Indonesian tourists visiting the Gili Islands	
		B1	Psychological	Information related to how the information provided by the chatbot would impact the users psychologically	"... the experience on Gili can be an extraordinary trigger for them to become [environmentally] aware" (S3, interview)
		B2	Financial	Information related to how the information provided by the chatbot would impact the users financially	"I am [from] a community with [limited] human resources and economic power. What is your deal [referring to the chatbot]? I'm uncomfortable. ... Why are you reminding me? Do you have any [tangible benefits] for me? On the other hand, I need [my private motorcycle]. If [the public transportation] is expensive, what should I ride? If you don't give me money, why do you ban me?" (D2, interview)
SPI			Spillover effects	The potential broader impacts that the technological innovation can bring beyond its main purpose	
C			On the government side	As the assumed investor, it can be expected that the government will also benefit from the broader positive impacts created by the technological innovation	
		C1	Enhanced tourism villages	Information related to how the chatbot can be used to enhance the existing Indonesian government's program called "Tourism Village"	"[this innovation] can help raise the name of Indonesia. So, it's like people are starting to pay more and more attention, we are the first to get this concept" (D8, interview)
		C2	Responsible repeat visit	Information related to the proposed idea of carbon offsetting scheme implementation for Gili tourists	"As long as there are rules and regulations that we can formalize, I don't think it's a problem and [as long as it] won't burden tourists. ... We [will] support the innovation [with respect to] the carbon [offsetting] ... as we will know that people coming here are already emission-free" (D4, interview)
		C3	Pro-sustainable businesses	Information related to how businesses should be encouraged to be pro-sustainable	"It would be good if this government initiative is supported by NGOs, ... local business-people ... restaurants A, B, C, and [customers can] get a discount. So, the win-win solution [is] in the [creation of an] ecosystem [for promoting sustainability]" (T3, interview)
		C4	Incremental improvement across different sectors	Information related to how the improvement caused by the chatbot can reach different sectors outside of tourism	"Let's say [the tourists on the Gili Islands are] teenagers and young adults ... when they have matured as individuals, they already have ... a positive attitude [toward the environment]. Project this to, say, 10 years, 20 years into the future, so ... from the side of human resources investment, [it] is extraordinary" (S4, interview)
		C5	New job opportunities	Information related to how technological innovation can potentially create new job opportunities	"When there is a conversion workshop [for electric vehicles], there is a derivative industry. There is a homemade battery industry run by electronics geeks in the city ... if you buy it, it's expensive—10 million [Indonesian Rupiah]. If assembled by them, it's only 3 million [Indonesian Rupiah]" (S1, interview)
	PRO		Pro-environmental behavior spillover	The behavior change of tourists as a result of using the technology and reflecting on their experience in Gili	"[the change in our behavior will] depend on the government. If that's what's implemented, we'll comply" (D3, interview)

(continued)

Appendix 3. (continued)

Level 1 Code	Level 2 Code	Level 3 Code	Code Name	Definition	Sample Quote
	SUS		Sustainable living in a developing country	Information related to how sustainable living ecosystem can be promoted in a developing country such as Indonesia	"We have to start investing in sustainable technologies for developing nations and pulling them off the ground. Then that's the kind of sustainable way for everybody ... that comes from needing better funding but also better education and school, and increasing awareness in general" (S6, interview)

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the Center for Higher Education Funding, Ministry of Education, Culture, Research and Technology of the Republic of Indonesia, and the Indonesian Endowment Fund for Education under the scholarship grant received by the corresponding author.

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Data Availability Statement

Data can be made available upon request.

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